

CITS3002: Lab 5

Exercise 1:

In an IEEE 802.11 wireless network, the bit error probability is p . A station uses the RTS/CTS mechanism prior to transmitting a data frame. The transmission process consists of four frames in the following order: RTS, CTS, Data, and ACK. Let the sizes of these frames be L_{RTS} , L_{CTS} , L_{Data} , and L_{ACK} , respectively. Assume that no collisions occur and that transmission errors occur only due to bit errors.

- Write the probability that each individual frame is received correctly.
- Write the probability that the entire transmission exchange (RTS, CTS, Data, and ACK) is completed successfully without any bit error in one attempt.
- If $p = 10^{-4}$, $L_{RTS} = 160$ bits, $L_{CTS} = 112$ bits, $L_{DATA} = 12144$ bits, and $L_{ACK} = 112$ bits, compute the numerical probability of success.

Exercise 2:

Write a Python program that implements Dijkstra's algorithm to find the shortest path from router A to every other node in the topology:

(A,B)=4, (A,C)=2, (B,C)=1, (B,D)=5, (C,D)=8, (C,E)=10, (D,E)=2, (D,F)=6, (E,F)=3

Exercise 3:

Assume node H is directly connected to nodes I, J, and K. The link costs from H to its neighbors are:

$$(H, I) = 2, \quad (H, J) = 3, \quad (H, K) = 6$$

At a certain time, node H receives the following distance vectors from its neighbors:

Vector of node I	
Dest	Cost
H	2
I	0
J	2
K	5
L	4
M	7

Vector of node J	
Dest	Cost
H	3
I	2
J	0
K	1
L	5
M	6

Vector of node K	
Dest	Cost
H	6
I	5
J	1
K	0
L	2
M	3

Using the distance vector routing, construct the routing table for node H.